

Lecture 1: Introduction to Memory Management

Fundamentals of Operating System Design (CEUC301)

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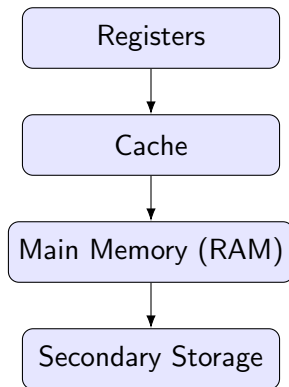
Learning Outcomes

- Explain the need for memory management.
- Describe the memory hierarchy.
- Explain the functions of memory management.
- Relate memory management to system performance.

Why Memory Management?

- Efficient utilization of main memory
- Support multiprogramming
- Protection between processes
- Controlled memory sharing
- Efficient allocation and deallocation
- Foundation for virtual memory

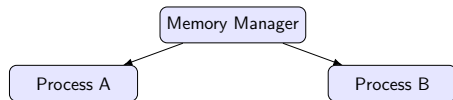
Memory Hierarchy



Speed decreases and capacity increases from top to bottom.

Functions of Memory Management

- Allocation and deallocation
- Address translation
- Protection
- Sharing
- Relocation
- Virtual memory support

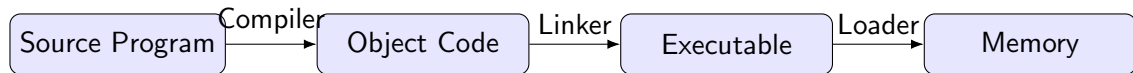


Address Binding

Address binding is the process of mapping program addresses to memory addresses.

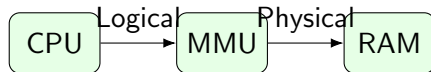
Binding Type	Description
Compile-time	Fixed memory location known before execution
Load-time	Address decided when program is loaded
Execution-time	Address translated during execution using MMU

Address Binding Process



Logical vs Physical Address

Logical Address	Generated by CPU
Physical Address	Actual RAM location
Visibility	Logical address is visible to program
Translation	Performed by MMU



Memory Management Unit (MMU)

The Memory Management Unit (MMU) is hardware that converts logical addresses into physical addresses.

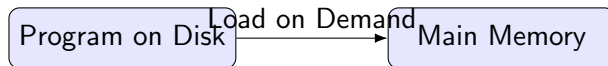
- Performs run-time address translation
- Enables relocation
- Supports memory protection
- Forms the basis of virtual memory

CPU $\longrightarrow$ MMU $\longrightarrow$ Main Memory

Definition

Dynamic loading loads a routine into memory only when it is first required.

- Unused routines remain on secondary storage.
- Saves main memory.
- Suitable for large applications.



Definition

Dynamic linking postpones linking of libraries until program execution.

Advantages

- Smaller executable files
- Shared libraries reduce memory usage
- Easy library updates

Examples

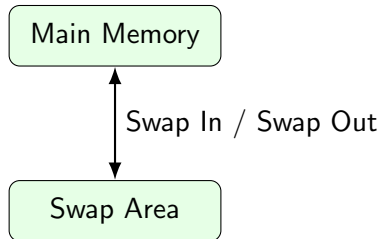
- Linux: .so libraries
- Windows: DLL files

Dynamic Loading vs Dynamic Linking

Feature	Dynamic Loading	Dynamic Linking
Purpose	Load program modules	Link shared libraries
When?	During execution	During execution
Object	Program routine	Library
Benefit	Saves memory	Reduces executable size
Example	Plug-in module	Shared libraries (.so/.dll)

Swapping

- Entire process is moved between RAM and disk.
- Frees memory for other processes.
- Improves multiprogramming.
- Used when RAM is insufficient.



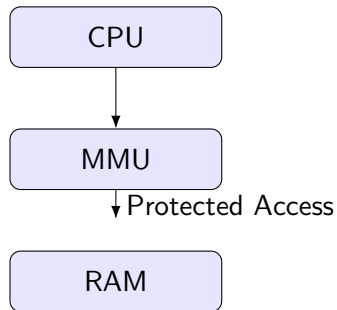
Memory Protection and Relocation

Memory Protection

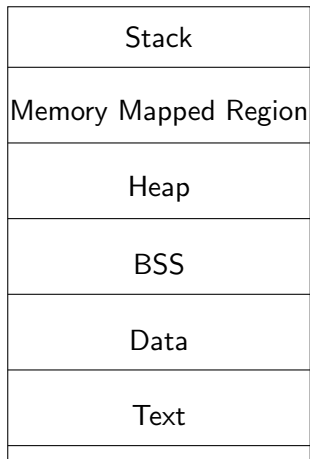
- Prevents unauthorized memory access
- Isolates processes
- Protects the operating system

Relocation

- Allows a program to execute from different memory locations
- Supported by execution-time address translation



Linux Process Memory Layout



Text → Data → BSS → Heap ↑ Stack ↓

Summary

- Need for memory management
- Memory hierarchy
- Functions of memory management
- Address binding
- Logical vs Physical address
- Memory Management Unit (MMU)
- Dynamic loading and dynamic linking
- Swapping
- Protection and relocation

Questions

- ① Why is memory management required?
- ② Explain the memory hierarchy.
- ③ Compare logical and physical addresses.
- ④ Explain the role of the MMU.
- ⑤ Differentiate dynamic loading and dynamic linking.
- ⑥ What is swapping?
- ⑦ Explain memory protection and relocation.